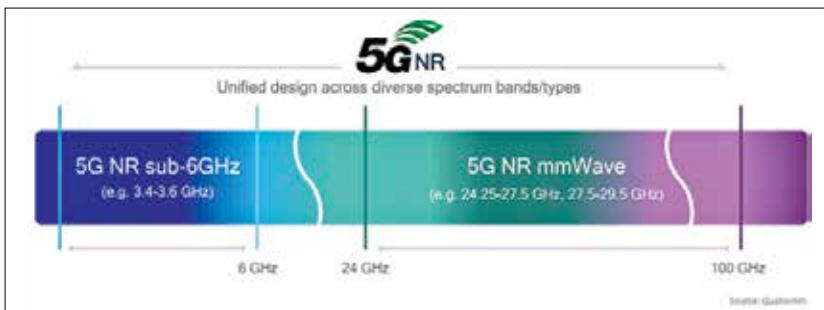




The 5G NR spectrum

customers. The 5G New Radio (NR) starts off from 600 MHz and scales all the way up to 4.7 GHz. Within it we find the low band and the mid-band spectrums. Major providers in the US such as T-Mobile have already started deploying hardware for the 600 MHz band. While this range starts off much lower than what existing technologies use, there is a significant overlap with 3G and 4G LTE bands. This low band will have a peak bandwidth of 100 Mbps. As of now, the most popular band in this sub-6 GHz spectrum is expected to be 3.5 GHz.

Then we have the mid-band spectrum which offers a peak bandwidth of 1 Gbps and is being adopted by major telecom providers as well. Since this mid-band operates at a higher frequency, it has lower penetration and coverage area. To overcome this, we will see the application of Massive MIMO.

Lastly, you have the high-band spectrum which is known as mmWave (millimeter Wave). This spectrum is greater than 6 GHz with

the most popular band being the 24.25 - 40 GHz spectrum. Within this, we have the bands that have been standardised. Right now we have four bands with the most popular one being the 24 GHz band. When we refer to 5G, we're often talking about this high-band spectrum because it is rare-



A small cell in Bangalore

ly used and thus presents a lot of challenges. Peak bandwidth in mmWave as per Release 15 of the 5G protocol is 10 Gbps but this is expected to jump to 20 Gbps by the time Release 16 is ratified. It goes without saying that at such high frequencies, you are going

to get absolutely horrible penetration. To counter this, smaller low power base nodes called Small Cells were developed. These Small Cells are the size of backpacks and consume very low power. Each Small Cell consists of hundreds of antennae and is supposed to be placed at regular distances from one another. This takes care of the coverage issues since these Small Cells are small enough to be strapped onto street light poles or any other utility pole for that matter. The unfortunate thing here is that you'll need to install thousands of these in an average city to ensure proper coverage.

Small Cells are SA (Standalone) devices which only support 5G networking. To make things easier for deployment, telecom operators will be relying on NSA (Non Standalone) devices which allow them to add 5G networking capabilities onto existing 4G hardware. This will, obviously, have limitations but it allows faster deployment without massive investment.

5G TIMELINE

CES and MWC 2019 saw a lot of 5G related announcements. The first client gateways were showcased earlier this year at CES 2019 and at MWC we saw hardware manufacturers such as Qualcomm, Ericsson and Intel come out with several product demonstration on their latest 5G solutions. We even saw the first 5G smartphones being announced in late 2018, albeit they were reference designs for manufacturers to base their models on. In fact, by the time you receive this magazine, you will have seen the world's first 5G commercial smartphone from a notable manufacturer.

As of now, 3GPP, the standards organisation which handles 5G protocol has not even finalised Release 15. Release 15 is the first release of the wireless protocol for 5G and all the things we see right now are based on the draft version that was released early in 2018. This release is happening over two phases and the final phase for Release 15 is expected to happen at the end of Q2 2019. This would be the one where the peak bandwidth is limited to 10 Gbps. Release 16, which

QAM

Quadrature Amplitude Modulation is the method that allows the combination of two amplitude modulated (AM) signals into a single channel. It isn't exactly like multiplexing but comes quite close. The two carrier waves are out of phase with each other by 90°, also known as orthogonality, and they're combined. Because they're out of phase in an orthogonal fashion, it becomes easy to separate the two analog signals once they're received. That's how you can pack two signals together in the same channel. This allows for much higher throughput and isn't a new technology. In fact, most of your TV signals and also the Wi-Fi in your homes make use of QAM to increase throughput. The 5G specification allows for 64-QAM or 256-QAM. 64-QAM allows using of 6-bits per symbol and 256-QAM allows the use of 8-bits per symbol. What that basically means is that the each shift in phase combined with the amplitude at that point allows the wave to indicate one digital value. So increasing the number of bits allows more values to be incorporated.